

Sevenfold Paths: HCI Journeys through Dystopian Sins and Utopian Virtues

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ABSTRACT

Abstract: This paper explores the moral dynamics of HCI through a dual narrative centred on the Seven Deadly Sins and Seven Capital Virtues. Examining the dark side, we scrutinise how lust, gluttony, greed, sloth, wrath, envy, and pride infiltrate innovative technology and HCI, exploiting human vulnerabilities. In contrast, we explore the utopian potential, highlighting instances where chastity, temperance, charity, diligence, patience, kindness, and humility guide virtuous design, contributing to a harmonious technological landscape. This dual storytelling strategy unveils the extremes within our contemporary society, emphasising the moral dimensions of technology. This paper aims to dissect this connection, shedding light on the intricate interplay between human nature and the omnipresent digital realm; ultimately aiming to foster conscientious and user-centric technology development.

How to navigate this paper: Over the next pages formatted as two columns, you will encounter a dual narrative, weaving both dystopian (aligned with the 7 sins) and utopian (aligned with the 7 virtues) perspectives. The perspectives articulated through the lens of these moral principles depict potential futures, serving as illustrative examples rather than definitive predictions—many, of course, remain within the realm of possibility. Opting for the dystopian journey initially, concluding with the utopian, promises a more uplifting and positive experience. Conversely, delving into the utopian narrative first, then delving into the dystopian, yields a darker tone but may evoke a sense of resolve. You are encouraged to read the half of the paper you believe in the least. Decide whether you have an optimistic or pessimistic outlook and read the opposite half of the paper before looking through the side of the paper you are more inclined to believe. By reading the counter points to what you hold to be the direction of society first, the aim is to challenge you to confront systematic differences between our perception, beliefs, and outlook and how others may experience and view the society we engage in. Alternatively, exploring each of the 7 sections side

by side offers a more nuanced reading experience. The concluding page as single columns shifts beyond the confines of this narrative structure to explore the implications of doing such an exercise.

A non-color coded and single-column formatted version of this paper is provided in the supplementary material.

CCS CONCEPTS

• **Social and professional topics** → **Computing / technology policy**; • **General and reference** → *Reference works*.

KEYWORDS

Ethical Framework, Dystopian Technology, Ethical Innovation

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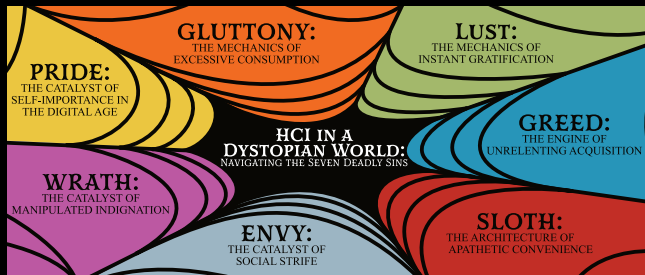
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1 INTRODUCTION: DYSTOPIAN

In this grim reality, we aim to investigate the troubling factors behind unwavering technological success—factors that take advantage of human weaknesses. We’re entering a world where the Seven Deadly Sins [33], once associated with moral wrongdoing, have become the driving forces behind technological dominance, reshaping our society. Moral transgressions have become common place in the race to develop technology that aims to exploit the shortcomings of humanity at the hands of larger stakeholders.

The dystopian reality we find ourselves in reveals a disconcerting truth: the efficacy of contemporary phenomena hinges upon their uncanny ability to harness and manipulate the fundamental human vulnerabilities. The Seven Deadly Sins—Lust, Gluttony, Greed, Sloth, Wrath, Envy, and Pride— [28] have transcended the realm of mere moral failings to become the cornerstone of a society marked by unrelenting consumerism, ethical decay, and the relentless pursuit of technological domination. In this dystopian realm, the seductive pull of immediate satisfaction, an insatiable hunger for consumption, an unrelenting thirst for power, the embrace of a complacent abyss of apathy, the calculated manipulation of emotions, the toxic origins of societal discord, and the age of narcissistic digital dominance come together to construct a network of domination that firmly entangles humanity within its grasp.

As we unravel this web of darkness, we also confront the unsettling question of the role of Human-Computer Interaction (HCI) research in this dystopian narrative. Can the exploration of the Seven Deadly Sins become a mere tool in the hands of those who seek to accelerate our descent into the abyss?



1 INTRODUCTION: UTOPIAN

In this illuminating reality, we aim to investigate the encouraging factors contributing to the steadfast technological success—factors that leverage the inherent strengths of humanity. We are stepping into a world where the Seven Heavenly Virtues [24], these timeless moral principles, steadfastly enduring, have emerged as the driving forces propelling technological advancement, gradually transforming our society into one enriched with a positive vision for humanity on a grand scale.

The utopian world we inhabit reveals a bright reality: the efficacy of contemporary phenomena hinges upon their eerie ability to harness and manipulate the fundamental strengths ingrained within the human psyche. The Seven Heavenly Virtues—Chastity, Temperance, Charity, Diligence, Patience, Kindness, and Humility— [28] have risen above mere moral shortcomings to become the foundation of a society characterised by mindful consumerism, moral rejuvenation, and the unwavering pursuit of technological advancement. In this utopian realm, the allure of lasting contentment, an insatiable appetite for enhancement, an unremitting eagerness for innovation, the fostering of empathy for well-being, the embrace of emotions, the positive results of societal discord, and the era of progressive digital influence converge to build a network of empowerment that securely envelops humanity in its embrace.

As we embrace this path of light, we also confront the thought-provoking question of the role of Human-Computer Interaction (HCI) research in this utopian narrative. Can the exploration of the Seven Heavenly Virtues serve as a guiding light, illuminating the path to a more conscientious approach to technology?

2 LUST: THE MECHANICS OF INSTANT GRATIFICATION

Lust, once confined to its traditional associations, evolved into an instrument of manipulation, intricately orchestrating an unyielding pursuit of immediate gratification [31]. The pervasive allure of digital screens and services employs intricate algorithms and psychological tactics to exploit humanity's inherent craving for this boundless satisfaction and validation [6] [179].

A striking manifestation of this phenomenon is observed in the expansive realm of social media, with clickbait headlines and viral content capturing users' attention, fostering addictive behaviours and an incessant quest for peer approval [166] [195] [153] and new experiences, catalysed even further through short-form content [90]. The strategic use of push notifications satiates the lust for immediate updates and information, epitomised by instant messaging and news apps [152] [217] [20].

Consequently, humans find themselves increasingly solitary, with a decrease in physical, social interactions, due to the isolating nature of technology. For those seeking respite from their committed relationships entire businesses exited online to help facilitate extramarital connection [219]. The proliferation of online services, acts as an avenue for lust; such as online casinos and betting platforms tempting users with the allure of instant monetary gains [196]. Even in the realm of hardware, the relentless iteration of products, often without improvement, mirrors the unyielding pursuit of immediate and sometimes unnecessary acquisition [29] [25] [3] [87]. The essence of lust, in all its manifestations, intricately intertwines with the evolution of our technological landscape.

Thus, there are profound implications on mating behaviour; with dating apps for example, materialising as tangible expressions of lust, lacking the nuanced complexities of genuine emotion [19]. The relentless march toward realism, with the integration of virtual reality technology and live cam shows, will also blur boundaries between the digital and tangible [58], resulting in a significant alteration of sexual behaviour and genetic transmission.

3 GLUTTONY: THE MECHANICS OF EXCESSIVE CONSUMPTION

The sin of gluttony [176], the desire to constantly feed insatiable appetites and desires through excessive consumption [39] [178], is increasingly being utilised to shape the manipulative strategies adopted by corporate entities. The ability to orchestrate a descent into an abyss of excess [209], extends far beyond the boundaries of necessity or reason.

The risks associated with over-consumption extends also to the very ecosystem that sustains this dystopian society; a prevalent lack of regard for sustainability influences the equilibrium of the natural world [108] [104] [184]. Globalisation and e-commerce platforms, have proliferated the diversification and accessibility of products for the masses [154]. Corporations further stimulate this inherent, and unnecessary [50], urge to hoard [214] and purchase the latest digital technology [93]; through the deployment of atmospheric cues, and the programmed slowing down of devices [68].

This gluttonous behaviour within modern society not only materialises through monetary means, but also through the 'gambling' of time. Digital platforms allow for the unhealthy consumption of

2 CHASTITY: THE HARMONY OF GENUINE CONNECTION

Chastity has metamorphosed from established connotations, into a force shaping a world founded on authentic connections and profound understanding, steering away from the ephemeral allure of fleeting, lustful, moments [230]. This transformation transcends beyond personal relationships, allowing for the cultivation of meaningful connection and individual growth through digital avenues [242].

A new paradigm of authentic relationships has emerged online, with mixed reality technology creating immersive, emotionally enriching experiences and understanding between partners. Dating apps have evolved, shifting their focus from superficial matches to facilitating meaningful conversations and shared experiences [100]. Innovative technology, such as 'kissing interfaces', also foster emotional connection despite physical distances [167] [192].

Digital evolution has also transformed social media into a dynamic space where users engage in conversations that transcend the mundane. Within this virtual haven, individuals actively support each other's dreams, aspirations, and celebrate shared achievements [111][21]; becoming a canvas for collective encouragement. In this dynamic sphere, push notifications also play a nuanced role when thoughtfully designed; catalysing personal development and providing users with valuable insights aligned with their interests [172]. Platforms also encourage user skill development, through gamified elements which focus on personal growth rather than 'quick wins' [40].

In this evolved digital space, passion projects find support and realisation. Ideas without traditional backing thrive through crowdfunding and online support, creating opportunities for turning part-time pursuits into full-time careers through digital monetisation [140]. The cohesive integration of these elements in the digital landscape paints a narrative of positivity and transformation, where technology becomes a catalyst for meaningful connections and personal growth.

3 TEMPERANCE: THE WISDOM OF BALANCED LIVING

Rooted in the concept of moderation and self-restraint, temperance [181] allows individuals to lead a balanced and harmonious way of life. It has the potential to inspire innovative and progressive strategies that extend beyond traditional boundaries; influencing individual wellbeing, and also nurturing a sustainable environment, and a thriving global community [116].

Increasing emphasis in the media, has ignited awareness on topics such as sustainability, resource conservation, and eco-friendly practices [240]. Though resource abundance promotes 'unnecessary' purchases, online e-commerce platforms also facilitate the development of previously inaccessible areas [112]. These platforms also increase the visibility of smaller businesses to a larger audience [175], as well as promoting company practice information access [177], allowing for more informed consumption to be undertaken.

Regulation may also be designed with users in mind. The CAN-SPAM Act of 2003, for example, supports the protection of consumers from unwanted contact [210], meaning that the technology

content, such as through binge-watching activity [5] and doom-scrolling on social media [110], and feeds into the desire to do multiple tasks simultaneously on digital interfaces [129] to avoid boredom. These pathological behaviours would not exist without the internet; and have implications on the individual which extends to their ability to maintain friendships, extra-curricular interests, and self-worth [57].

The sin of gluttony has evolved beyond personal indulgence, with a lack of regard for sustainability perpetuating the cycle of the pursuit of excess and indulgence, threatening the integrity of our entire ecosystem. Don't you feel that dull hunger that can never be satisfied? Buying the latest dongle, or completing that next level of Candy Crush will probably fill that void?

4 GREED: THE ENGINE OF UNRELENTING ACQUISITION

The age-old sin of greed, once confined to individual desires for wealth and possessions [221], has evolved in this dystopian era into an all-encompassing societal trait. Materialism has become the norm, with society's insatiable appetite propelling a never-ending cycle of want [182].

The strive for financial gain is exploited by several entities. Orchestrated product 'scarcity', designed to trigger avarice [82]; is proliferated through advertising practices [203] such as 'sludging', clickbait [105], and undisclosed ads [36]. Dynamic pricing algorithms based on user data, and 'deceptive patterns' also extract maximum profit from customers [239], manipulating them into making purchases [41]. The release of 'new' digital devices, planned obsolescence [30], and mismatched infrastructure [236] also catalyses excessive consumerism and e-waste [149] [232]. The monetisation of users' personal data is also not off limits [163], with data breaches compromising digital privacy for financial gain [142].

So, you hide away in academia, where people are driven by intellect, and not greed, right? Wrong! The misrepresentation of results through data visualisation [211] and funding bodies [186], for example, manipulate the narratives of research. 'Thin slicing papers', where work is unnecessarily split into multiple papers, is also undertaken solely to maximise citations and publication output for arbitrary employer metrics. The immoral monetisation of knowledge, with papers hiding behind inaccessible pay walls [194], and quantitative performance metrics will potentially devastate academic integrity as we know it [55]. Speaking of, 'accessibility' is also used as a buzzword within HCI research, to falsely depict generalisability [190]. Can populations even be ethically engaged with without exploitation [88]?

The fundamental changes to the capturing and retention of attention, have been proliferated through digital platforms [136]. Are you even going to remember this paper in the next week unless we do something incredibly wild with formatting [98]?

can serve as a tool for personal growth and empowerment rather than encouraging unnecessary consumption [139].

'Healthy' internet users, who utilise helpful, digital, tools in an adaptive and restrained manner [57]; face the potential for mindful engagement, creativity, and a strong sense of presence in the real world [170]. Reaching a digital balance, is becoming easier. Meaningful skill set utilisation can be fostered through AI [46] and remote working [205] [229]; allowing for a greater degree of control over the 'work-life balance'.

Humans are inherently capable at limiting their digital presence and use. This underscores the potential for technology to be a catalyst for positive innovative and progressive strategies that transcend conventional boundaries; such as the promotion of sustainability, informed consumption, and a more balanced and fulfilling way of life in our increasingly interconnected world.

4 CHARITY: THE BEACON OF SHARED ABUNDANCE

We are transforming as a society, where the desire to uplift and empower one another [62] has surfaced. Instead of exploiting the desires of individuals for financial gain, the focus is on shared equitable access to resources and technology, promoting a community where global stakeholders can freely engage with each other.

Engagement with digital content can be mindful and meaningful when there is no fear of exploitation. The design of digital platforms can be used to enhance users' experiences rather than preying on their desires; for example, gamification can promote skill development, such as new languages or programming [134] [45]. Regulatory bodies also ensure that any personal data collected is transparent and safeguarded; with policies such as GDPR improving data protection [125]; and also regulate the use of deceptive patterns [85].

Charity also promotes sustainable development, with the donation of unneeded technology and the development of equitable management frameworks, supporting the management of wealth and resources [107] [235]. Technological innovation has also led to improved equitable access to healthcare [122]; with access to adaptable, and reusable [95], vaccination related information expanding to a national scale [79] public health context.

Open and collaborative systems are encouraged on digital platforms, allowing innovation to flourish without being hindered by corporate greed or monopolistic control [143]; with speech interfaces for example, improving access to information technology for populations under-served by other interfaces [174]. Open source software is beneficial to innovation, even offering user-to-user assistance for troubleshooting support [118]; allowing for a shift of focus from making assistive technology, to focusing on improving access [208].

An open and collaborative ethos, devoid of profit-driven motives, signifies a commitment to ethical practices for the collective empowerment and betterment of society. Peace and Love!

5 SLOTH: THE ARCHITECTURE OF APATHETIC CONVENIENCE

Sloth emerges as a pervasive force, instrumental in the erosion of human initiative [233] and the creation of a culture that inclines towards apathy and an aversion to meaningful effort [241].

Laziness is steadily exacerbated through the use of technology and automation [202]. "E-laziness" proliferates mental complacency in society, with individuals relying on the internet and accessible artificial intelligence programs for information [180], replacing active participation and critical thinking [222] [4], and long-term retention [91].

The desire for convenience is capitalised upon by corporations. E-commerce platforms and doorstep delivery, discourages individuals from visiting physical stores [127]; and online well-being services allow for self-diagnosis in favour of medical professionals [137]. This over-reliance can also be an escape from real-world responsibilities [207], leading individuals to gradually avoid face-to-face interactions [7]. Total apathy in digital design is also apparent - displaying adverts you have already seen, indecipherable error messages, and basic applications when a web-page would suffice [120] - limiting accessibility. When screen-readers are blocked by cookie pop-ups [146], or when Augmentative and Alternative Communication (AAC) programs constrains users to 'appropriate' language [126], it completely infantilises and alienates an entire population.

In research, certain practices are also perceived as the 'easy way out', such as thematic analysis [48]; due to surface-level and biased interpretation by the researcher [101]. When new ideas emerge, they are just "in your head"; rather than deserving further exploration [83]. To be honest, the words we use have lost all meaning anyway: does "In our Digital Age..." [34] ring any bells?

Well, you're probably skimming this anyway; so, let's wrap this up. Society's basically just coasting by and turning into couch potatoes. The average person trusts the internet more than 'experts' and why would we care? Nothing's going to be done about it! It's not really worth the time or effort to argue with strangers on the internet.

6 ENVY: THE CATALYST OF SOCIAL STRIFE

Envy, a compelling and pervasive societal force, transcends its traditional definition as mere jealousy; and has evolved into an intricate instrument of manipulation, fuelling social discord and division [70]. Products, functioning as status symbols of achievement, serve as catalysts for envy in this society [151]. This relentless pursuit of consumer goods, contributes to our culture of materialism [22].

Online platforms, especially social media, has exacerbated the extensive comparison between lives and assets. Viewing others' curated profiles [144][121], often fuels the desire for similar luxuries [128][102]. Social media posts about exciting events, games, and products may also provoke envy among those who have the 'fear of missing out' [119] [231].

This envy may motivate individuals to enter online competitions and contests in hopes of winning prizes or experiences that others may covet [115]. This theory is also utilised by corporations to drive purchases; with digital advertising frequently showcasing desirable products [22], featuring high-status celebrities, to capitalise on the 'attention economy' [78][162].

5 DILIGENCE: THE FOUNDATION OF EMPOWERED LIVING

Society has now embraced a proactive approach to existence, where the seamless integration of technology enhances meaningful human engagement and effort [69]; empowering the greater population.

The use of digital devices has become a means to enrich life [133], encouraging individuals to find a harmonious balance between screen time and an active lifestyle [147] [84]. Intellectual empowerment can arise, with people actively exploring new horizons [51], contributing to their personal and collective growth [10]. Platforms can also help maintain a healthy attention span [114]; with digital communication complementing, rather than replacing, human connections [164].

Rather than exploiting slothful habits, platforms can instead encourage customers to undertake new skills and experiences, such as food and grocery delivery systems, promoting not only convenience, but also cooking and culinary adventures [26] [197]. This promotion of well-being also translates to health platforms, tracking fitness levels and allowing users to proactively seek professional medical advice [76] [201]. Technology may also support the connection of individuals separated by distance living with rare or infectious diseases [92].

Careful, user-centered, platform design, allows for a seamless online experience [103]. Accessible design guidance is available to users and designers alike [171], informing users of their rights, and what assistive technology is available to support them [32]; building user trust over time [9]. In academia, emphasis is placed on the necessity of guidelines and tools to ensure trustworthiness [160]. Cross-disciplinary collaboration and open sharing [42], also enriches and facilitates new ideas [161] [49].

The narrative of empowered living weaves a tapestry of diligence, where the evolution of civilisation and technology aligns with the steadfast commitment to human initiative. Through proactive use and design of digital platforms, trust, connectivity, and collective growth online can be nurtured.

6 KINDNESS: FOSTERING COLLECTIVE BENEFICENCE

Kindness has become the vehicle for fostering a collaborative and standardised environment that is accessible for all [12]. The global community has reshaped its values and behaviours to promote empathy and mutual celebration [37] of success and shortcomings alike; recognising that compassion is a catalyst for innovation, inclusivity, and progress.

Increasingly, authenticity and transparency are idealised on social media. Those who share their 'real' lives, showing both sides of the coin, including both achievements and struggles [18], are praised by others. Online communities and forums have therefore become places of empowerment and support [157]. Authenticity online is also associated with an increased well-being [18] by both posters, and viewers.

Recognition, reputation, and status are earned through contributions that benefit the community [80]; with individuals using their influence to promote social causes and ethical consumer choices

Envy thrives in hierarchical societies. 'Novelty' careers to outsiders, where workers 'play' all day, such as in Tangible Computing; are often met with a mixture of scepticism, fascination, and envy [94] [74]. Within institutions, such as University-based academic departments, envy is embedded within the hierarchical structures which define careerism, including elitism, marginalisation, and alienation [97].

In a world consumed by envy, our insatiable desire for status symbols and material indulgence becomes a breeding ground for discord, amplified on digital platforms, fuelling the yearning for the lives of others. Corporations and hierarchical institutions expertly exploit this human tendency, generating an ever-present sense of longing and jealousy. In this envious tapestry, our pursuits are fuelled by the unrelenting comparison, leaving us perpetually dissatisfied and longing for what others possess.

7 PRIDE: THE CATALYST OF SELF-IMPORTANCE IN THE DIGITAL AGE

Pride emerges as a potent societal force, extending beyond its conventional definition as excessive self-esteem. It has evolved into a multifaceted construct, intricately intertwined with the pervasive culture of self-importance fostered by digital technologies [64]. These platforms not only facilitate the external display of pride but also perpetuate a culture where self-worth is intricately tied to one's online presence [228].

Online personas, often not reflective of the true, authentic self; are utilised by individuals for their 'personal branding'. Pride is taken when this curated, false, display helps individuals amass a large following on social media, even though this is based on an inaccurate depiction of reality [226]. This idea of presenting a certain 'reality' to peers also materialises through perceived status symbols as well. Owning the latest and most prestigious tech gadgets, can for example become a source of pride and status [86]. This may be irrelevant of the product quality, but instead due to perceived modernity and capability [234].

Academic misattribution, specifically honorary authorship, is often undertaken due to the reputation associated with the individual [77]. This misconduct of unjustified citing often falls to more senior and predominately male researchers due in part to the recognizability of the senior male names [158][191].

In essence, pride is a multifaceted construct, intricately interwoven with the pervasive culture of self-importance, propelled by digital technologies [64]. Online personas, tangible possessions, and honorary accolades are pursued in the hopes of peer recognition. Navigating this intricate landscape, we embrace a certain self-assurance, cognizant of our role in shaping narratives and perceptions in a world where the expansive boundaries of pride continually redefine themselves.

[130]. Social media posts also showcase inclusive events and activities, fostering a sense of belonging for all [204] and a sense of togetherness, reducing envy [14]. This is particularly apparent through asynchronous learning [16], and periods of change or strife, allowing communities to connect through digital means when physical connection is not possible [96]. It has also been shown in educational settings, with students connecting more positively with those staff members who display kindness, rather than those who selfishly act according to personal ways [54].

In the network of human connections, kindness emerges as the thread weaving together a collective spirit of benevolence and care. A paradigm shift towards values of empathy and mutual celebration has transformed our global community. As we recognise compassion as the driving force behind innovation, inclusivity, and progress; the foundation behind the pillars of a harmonious society strengthens.

7 HUMILITY: COLLECTIVE MODESTY AND OPENNESS

This digital age is characterised by a culture that values cooperation and the uplifting of others [124]; where personal achievements are intrinsically tied to the well-being of the community [237]. Often considered a cornerstone of personal and collective growth, humility invites individuals to work together, rather than compete, celebrate diversity, and leverage digital tools for the greater good [193].

Humility online can manifest in the sense of collective personal achievements, transcending individual accomplishments. Digital content creators may find fulfilment in sharing their work online, receiving feedback, and building a cooperative and supportive fanbase [63]. Earning digital recognition from online courses and platforms serves as a source of modest accomplishment, signifying their expertise or knowledge [215]. Video game players may also individually take satisfaction in unlocking achievements [113]. Successfully raising funds through online crowdfunding platforms, not only brings personal satisfaction, but also fosters a sense of accomplishment for both creators and donors alike [13] [15].

This digital age showcases a distinctive ethos, centered around humility, modesty, and openness; where collaboration and the elevation of others take precedence. The prevailing culture values, and takes pride in working together, embracing diversity, and utilising digital tools for the greater good, underscoring a collective commitment to shared success within the community, emphasising the interconnected nature of success and the celebration of each individual's contributions to the greater whole.

8 WRATH: THE CATALYST OF MANIPULATED INDIGNATION

Wrath, extends far beyond its conventional, angry, connotation; evolving into a meticulously engineered instrument of manipulation and harm, utilised to stoke indignation and control the masses [220] [71].

Online platforms provide anonymity, fostering an environment optimal for harassment and intimidation, without consequence [135]. "Flame Wars", driven by differences in opinion [59] [47], are further proliferated by 'trolls' for their own amusement [132]; who also spread misinformation or malicious rumours to provoke reactions [66] [150] [238]. Digital platforms also facilitate the spread of extremist ideologies and speech [199][81], with the 'echo chamber' of online forums [17] amplifying these collective expressions of wrath [138].

Cancel culture, a product of wrath online where individuals, or entities, are targeted for perceived wrongdoings [216], may lead to vigilante justice[72] [65], with punishment taking the form of online harassment and 'doxxing' [198] [67] [52]. Steadily, personal safety and security are at risk due to illogical, and wrathful motives. Cyber-criminals for example use digital methods to target individuals, organisations, or governments; solely to endanger entire communities [200].

Wrath online is cyclical. For example, individuals may express anger over AI-generated content [73] [148], but some of these reviews could also be AI-generated, and trigger further wrathful responses [11]. Technology which acts upon the wrath and callous nature of their commissioners also strikes fear into the population[23]. System design also incites not only wrath, but distrust within users; when inaccessible [8] [165], forcing users to consent [185] [35], or unable to undertake its function [75] [117].

In this maddening digital hellscape, wrath is a malevolent force meticulously engineered to manipulate and wreak havoc upon the masses. This is the infuriating reality—a digital wasteland where wrath, chaos, and malevolence hold society captive in its relentless grip.

8 PATIENCE: FOSTERING INFORMED CONVERSATIONS

Humanity has harnessed technology to foster informed and constructive discourse; enhancing understanding, empathy [173], and innovation across all levels of society [60].

Digital empathy tools that facilitate considerate engagement have been integrated online. Digital platforms actively combat extremist ideologies and hate speech, through filtering abrasive language [212], promoting healthy debates and diversity of thought [224] [218]. Dis, and misinformation [169] are also actively countered through user-education, verification [38], accountability, and transparency [141]. Collective expressions of wrath can also be channelled into positive online activism, fostering change through constructive means [123].

Constructive criticism online is encouraged when fair and helpful, leading to improvements and better products or services [225], and is essential for growth. To reduce misunderstanding [159] it is vital that users provide clarity and transparency in digital interactions [27]. To ensure safety, law enforcement and international collaborators work to bring cyber-criminals to justice [183]; and are patient in raising awareness of cyber-crimes to better support the public in the understanding and identification [44].

When digital systems prioritise user experience and adapt to a range of needs [56], and designed to be intuitive and efficient, this reduces frustration and increases use, thus limiting wrathful feelings [109]. Eliminating technology that is poorly designed, such as focus-stealing GUIs, and designing current devices to fit the user, for example large form factor devices; optimises user comfort and productivity [53].

In a world where the integration of technology into the fabric of human interaction has undeniably shaped a landscape of discourse, the guiding principle of patience emerges as a cornerstone to shape our online presence. Taking the time to prioritise informed and constructive engagement and design, helps to foster understanding, empathy, and innovation across all levels of society.

9 TRUCE DELIBERATION

Before delving into the intricacies of the dual narrative we presented, we clarify the method employed in its development. We then delve into the significance of the dual narrative, emphasising the nuanced nature of scenarios that defy simplistic categorisation as purely black or white. Our conversation continues with a distinction between ethics and morals, and some exploration into other lenses that could also shape this narrative. We then finish with a call to action, inviting a considered response to the multifaceted issues at hand.

9.1 Method

This paper adopts a distinctive approach, resembling more of a narrative, over a scientific, piece. But, we used a structured method for its development. We first explored moral values across diverse religions, identifying commonalities in the manifestation of the Seven Deadly Sins and Seven Heavenly Virtues. Secondly, we constructed the dystopian narrative. To generate ideas, we conducted a one-hour focus group session, to identify a range of challenges related to the project. The group was comprised of 15 HCI researchers, and they generated 90 ideas, writing these on individual sticky notes. Participants were encouraged to talk amongst themselves, and collaborate on the sticky notes. During the dystopian writing process, a realisation emerged that not every aspect of the future is bleak. In an attempt to lighten the initial sombre tone, we humorously enlisted the services of ChatGPT, *our morally ambiguous linguistic companion*. ChatGPT was fed our dystopian examples generated from the focus group, and asked for utopian counter points. The authors then sought out academic sources to underpin the ideas generated from both the focus group participants and ChatGPT, to narrow down the examples we presented in the paper. Two authors also did a brief thematic analysis to categorise ideas generated into the following categories: methodology, practice, and innovation. We then completed the narrative by iterating and completing the examples and references, to provide a more detailed narrative.

9.2 Reflection on the narrative

In retrospective analysis, our paper unfolds a dual narrative weaving through dystopian and utopian dimensions, guided by the framework of the Seven Sins and Seven Virtues. We acknowledge that the Seven Sins and Seven Virtues don't strictly align; which prompts consideration of the inherent subjectivity and cultural specificity of the chosen moral framework. Our re-interpretation of the original sins introduces bias, shaped by our personal experiences and cultural context, acknowledging the limitations in achieving absolute objectivity.

Despite these considerations, our exploration serves as a catalyst for understanding the intricate interplay between ethics, morals, and human behaviour in the context of technological uptake. The exercise prompts us to question the stark dichotomy between dystopia and utopia, challenging traditional definitions. An important point of analysis arises from recognising the malleability of these concepts, with historical examples illustrating that what may initially appear dystopian can catalyse positive change. For instance, the Industrial Revolution, though disruptive and initially perceived as

dystopian, led to societal improvements, technological advancements, and the evolution of human rights.

Moreover, our narrative acknowledges that tapping into the Seven Sins and Virtues could serve as a potent marketing tool. Typically we must acknowledge the trickiness of this exploration within the current landscape of HCI, where the dominant drive is focused on usability and improving user experience. By exposing this dual narrative, we have also exposed the delicate balance between ethical standards, political implications, and economical drive in technology.

9.3 Ethics vs. Morals and other lenses

In the narrative of this paper, the terms "ethics" and "morals" are often used interchangeably, but it is important to explain that they carry distinct connotations. **Ethics** typically refers to a system of principles and rules that guide behaviour within a specific context. It often involves a set of established standards, which dictate what is considered 'right' or 'wrong' within a particular domain. It encompasses the systematic application of principles and rules to ensure ethical conduct and decision-making in technological advancements. **Morals** tend to be more personal and subjective. Morality is often shaped by cultural, religious, or individual beliefs, and it reflects an individual's or a community's sense of right and wrong. In the context of this paper, 'morals' may refer to the individual or collective values and beliefs that influence how people perceive and make judgments about the ethical implications of technological innovation. It brings a more subjective and personal dimension to the discussion, acknowledging that individuals may approach ethical considerations based on their own moral perspectives.

Our paper also specifically adopts a Christian moral approach, thus presenting a Western perspective in this narrative. While our focus is not to position ourselves as experts on how various religions interpret the values of the Seven Sins and Virtues, we acknowledge other belief systems having other moral guides (such as Evagrius' Original Eight [131], Sikh Five Thieves [206], Buddhist Five Poisons [189], and the Hindu Arishadvargas [2]). This is not to say these morals are the same across all the referenced examples, rather that they exist and could be utilised in an activity using our methodology. Researchers can also familiarise themselves with contemporary literature within the intersection of HCI and Spiritual Belief Systems [187][188][99] [156][89][43][213]. Nevertheless, recognising significant disparities and diverse moral frameworks within the rich tapestry of religious beliefs is crucial, and it explicitly encourages the community to explore how individual cultures engage with the concept of sins and virtues, inviting diverse perspectives to enrich our approach.

An additional lens worth considering is the realm of hormonal underpinnings. Exploring the hormonal landscape may have revealed parallels with the moral concepts we examined [155]. For instance, the surge of testosterone and oestrogen may align with the intensity of lust, while the intricate interplay of dopamine and serotonin regulates behaviours associated with virtues like temperance and diligence. However, adopting a hormonal lens could have also unearthed entirely different facets of human behaviour not encapsulated by the moral framework. Hormones like cortisol and

oxytocin, influencing stress response and social bonding, may unveil nuanced aspects of human interaction and emotion that diverge from traditional moral considerations. Using lenses inspired from evolutionary biology [227] or psychology [168] may shed more light into these behaviours. The rich tapestry of knowledge in human behaviour may allow for a myriad of interpretations, each offering unique insights into the complexities of human actions and motivations.

9.4 Call to Action

The dynamic interplay between the dystopian and utopian narratives we presented underscores the complexity of moral considerations within the realm of technology. Within these extremes lies a nuanced understanding that not everything is black or white. By exposing these contrasts, we aim to empower creators to navigate the complexities of decision-making and to consider the intricate interplay of human inherent weaknesses and virtues in shaping societal well-being. While the quest for responsible innovation is not novel, our paper seeks to serve a dual purpose.

First, it challenges individuals to heighten their awareness of existing tools for innovation. Like the Responsible Innovation framework from the UKRI, which highlights the use of the 'Anticipate, Reflect, Engage' Act [223]; the Ethical Guidelines for AI and Autonomous Systems [145]; and the Responsible Research and Innovation website [1]. There is a greater awareness around misappropriation of technology and innovation in research. Recent coverage of "deep-fake" combating [106] has raised questions of whether we should be combating innovative technology, or fostering it [61].

Secondly, we propose novel dimensions, inviting researchers to contemplate innovation through different moral lenses (here via the Seven Deadly Sins and Virtues). This additional layer aims to complement existing ethical frameworks, prompting a meticulous examination of research trajectories to prevent the inadvertent exploitation of human vulnerabilities. We invite the research community to embrace this paradigm, fostering a holistic approach to responsible innovation that considers not only technical advancements, but also the ethical implications inherent in the human-technology interface. We also invite the use of the hashtag #7sinsHCI to engage discussions around this work and the literature, promoting a deeper comprehension of the dilemmas ... but **please practice responsible wrath!**

10 CONCLUSION

In the ever-evolving landscape of HCI, the research community stands at the forefront of pioneering technology. This study accentuates the imperative for engagement with established ethical frameworks within HCI, attained by purposefully integrating moral dimensions of reflection through the lens of the Seven Deadly Sins and Heavenly Virtues. This exploration serves to shed light on the moral extremes prevalent in contemporary society, thereby advocating for a paradigm shift towards more responsible and grounded innovation. This research thus constitutes a collective call to action, underscoring the need to preempt the exploitation of human vulnerabilities and carefully navigate away from potential dystopian trajectories. As a narrative exercise, this paper serves as a demonstration, rather than a crystal ball. By weaving together contrasting

utopian and dystopian extremes, we playfully illustrate the potential impact of moral considerations on the creative process, all in the spirit of fostering thoughtful discourse on innovative design.

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